

2.2 ELECTROLYTIC CAPACITORS

MIKROELEKTRONIKA

Electrolytic capacitors represent the special type of capacitors with fixed capacity value. Thanks to special construction, they can have exceptionally high capacity, ranging from one to several thousand μF . They are most frequently used in circuits for filtering, however they also have other purposes.

Electrolytic capacitors are polarized components, meaning they have positive and negative leads, which is very important when connecting it to a circuit. The positive lead or pin has to be connected to the point with a higher positive voltage than the negative lead. If it is connected in reverse the insulating layer inside the capacitor will be “dissolved” and the capacitor will be permanently damaged.

Explosion may also occur if capacitor is connected to voltage that exceeds its working voltage. In order to prevent such instances, one of the capacitor's connectors is very clearly marked with a + or -, while the working voltage is printed on the case.

Several models of electrolytic capacitors, as well as their symbols, are shown on the picture below.

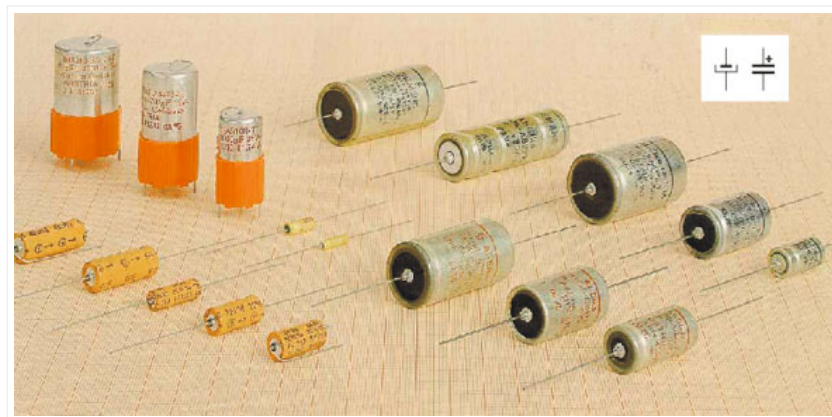


Fig. 2.4: Electrolytic capacitors

Tantalum capacitors represent a special type of electrolytic capacitor. Their parasitic inductance is much lower than standard aluminum electrolytic capacitors so that tantalum capacitors with significantly (even ten times) lower capacity can completely substitute an aluminum electrolytic capacitor.



